

Accelerating ionizable lipid discovery for mRNA delivery using machine learning and combinatorial chemistry

In the format provided by the
authors and unedited

Supplementary Information

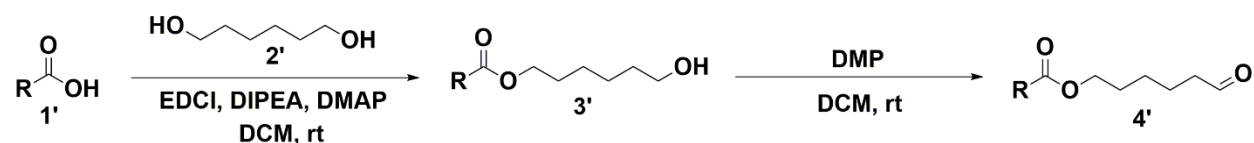
Table of contents

Lipid Synthesis.....	1
Supplementary Figures.....	12
Source Code.....	24

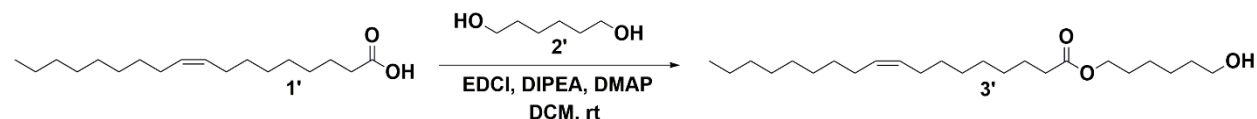
Lipid Synthesis

All solvents and reagents used were obtained commercially and used as such unless noted otherwise. ^1H and ^{13}C NMR spectra were recorded in CDCl_3 at room temperature using a Bruker Ultrashield 400 MHz instrument. Spectra were processed with Chemical shifts reported as parts per million (ppm) relative to TMS (0.00) for ^1H . Silica gel chromatography was performed on ISCO CombiFlash Rf+ Lumen Instruments using ISCO RediSep Rf Gold Flash Cartridges (particle size: 20-40 microns). All final compounds were confirmed by mass spectrometry using direct injection method on QTOF-HRMS (Agilent) coupled to an Agilent Infinity 1260 LC system. Only top performing lipids after screening by batch analysis were fully characterized by ^1H and ^{13}C NMR.

General synthesis of Tail 1



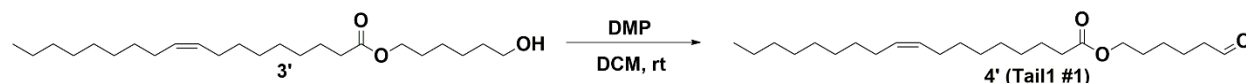
Synthesis of 6-hydroxyhexyl oleate, 3'



Oleic acid (5g, 17.7 mmol), *N*-(3-dimethylaminopropyl)-*N'*-ethylcarbodiimide hydrochloride (EDC HCl, 5.1g, 26.6 mmol), hexane-1,6-diol (10.5g, 88.5 mmol), 4-(dimethylamino) pyridine (DMAP, 1.1g, 8.9 mmol) and *N,N*-diisopropylethylamine (DIPEA, 6.2 ml, 35.4 mmol) were dissolved in dichloromethane (200 ml). The reaction was stirred at room temperature under nitrogen for 18 h, then washed with a saturated aqueous sodium bicarbonate

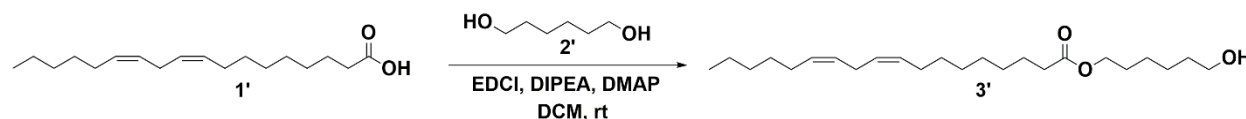
solution. The organic layer was separated, washed with brine, dried over Na₂SO₄, filtered, and the filtrate was evaporated under vacuum. The residue was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give 6-hydroxyhexyl oleate, **3'** (5.8g, 15.2 mmol, 86%) as a colorless oil. ¹H NMR (400 MHz, CDCl₃) δ 5.43 – 5.29 (m, 2H), 4.09 (t, *J* = 6.7 Hz, 2H), 3.67 (t, *J* = 6.5 Hz, 2H), 2.31 (t, *J* = 7.5 Hz, 2H), 2.03 (q, *J* = 6.5 Hz, 4H), 1.73 – 1.53 (m, 6H), 1.49 – 1.17 (m, 24H), 0.95 – 0.83 (m, 3H).

Synthesis of 6-oxohexyl oleate, **4'**



6-hydroxyhexyl oleate, **10** (5.8g, 15.2 mmol) was dissolved in dichloromethane 300 ml followed by Dess-Martin Periodinane (9.6g, 22.8 mmol). The reaction was stirred under nitrogen at room temperature for 2 h. After confirmation of reaction completion by thin layer chromatography, sodium thiosulfate pentahydrate (50% w/v, 200 ml) was added to the reaction and left to stir for additional 15 mins. Organic layer was thereafter separated, washed with brine, dried over Na₂SO₄, filtered, and the filtrate was evaporated under vacuum. The residue was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give 6-oxohexyl oleate, **4'** (3.5g, 9.2 mmol, 61%) as a colorless oil. ¹H NMR (400 MHz, CDCl₃) δ 9.79 (s, 1H), 5.42 – 5.33 (m, 2H), 4.09 (t, *J* = 6.6 Hz, 2H), 2.47 (td, *J* = 7.3, 1.7 Hz, 2H), 2.31 (t, *J* = 7.5 Hz, 2H), 2.03 (q, *J* = 6.5 Hz, 4H), 1.67 (dp, *J* = 15.1, 7.8 Hz, 7H), 1.48 – 1.21 (m, 23H), 0.90 (t, *J* = 6.8 Hz, 3H).

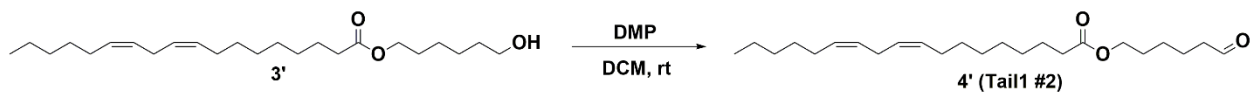
Synthesis of 6-hydroxyhexyl (9Z,12Z)-octadeca-9,12-dienoate, **3'**



6-hydroxyhexyl (9Z,12Z)-octadeca-9,12-dienoate, **3'** (linoleic acid) was made using synthesis of 6-hydroxyhexyl oleate, **3'** above.

¹H NMR (500 MHz, CDCl₃) δ 5.47 – 5.26 (m, 4H), 4.09 (t, *J* = 6.7 Hz, 2H), 3.67 (t, *J* = 6.5 Hz, 2H), 2.80 (t, *J* = 6.7 Hz, 2H), 2.31 (t, *J* = 7.6 Hz, 2H), 2.07 (q, *J* = 7.0 Hz, 4H), 1.74 – 1.53 (m, 7H), 1.49 – 1.18 (m, 18H), 0.91 (t, *J* = 6.8 Hz, 3H).

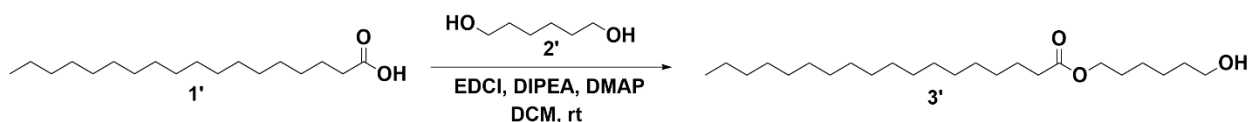
Synthesis of 6-oxohexyl (9Z,12Z)-octadeca-9,12-dienoate, **4'**



6-oxohexyl (9Z,12Z)-octadeca-9,12-dienoate, **4'** (linoleic acid) was made using synthesis of 6-oxohexyl oleate, **4'** above.

^1H NMR (400 MHz, CDCl_3) δ 9.79 (s, 1H), 5.59 – 5.15 (m, 4H), 4.09 (t, $J = 6.6$ Hz, 2H), 2.79 (t, $J = 6.4$ Hz, 2H), 2.31 (t, $J = 7.6$ Hz, 2H), 2.06 (dd, $J = 8.1, 5.7$ Hz, 4H), 1.68 (dq, $J = 14.4, 7.2$ Hz, 8H), 1.44 – 1.22 (m, 16H), 0.91 (t, $J = 6.7$ Hz, 3H).

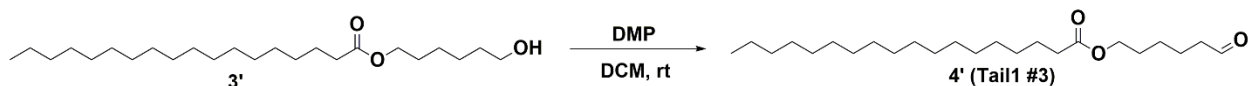
Synthesis of 6-hydroxyhexyl stearate, **3'**



6-hydroxyhexyl stearate **3'** (octadecanoic acid) was made using synthesis of 6-hydroxyhexyl oleate, **3'** above.

^1H NMR (500 MHz, CDCl_3) δ 4.09 (t, $J = 6.7$ Hz, 2H), 3.67 (t, $J = 6.6$ Hz, 2H), 2.31 (t, $J = 7.6$ Hz, 2H), 1.64 (ddt, $J = 22.0, 14.5, 7.1$ Hz, 7H), 1.42 (dq, $J = 6.6, 2.9$ Hz, 4H), 1.28 (d, $J = 2.5$ Hz, 27H), 0.90 (t, $J = 6.9$ Hz, 3H).

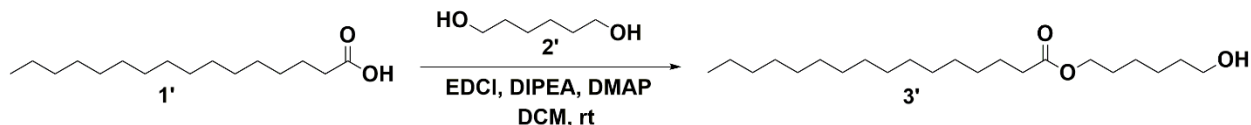
Synthesis of 6-oxohexyl stearate, **4'**



6-oxohexyl stearate **4'** (octadecanoic acid) was made using synthesis of 6-oxohexyl oleate, **4'** above.

^1H NMR (400 MHz, CDCl_3) δ 9.80 (s, 1H), 4.09 (t, $J = 6.6$ Hz, 2H), 2.48 (td, $J = 7.3, 1.7$ Hz, 2H), 2.31 (t, $J = 7.6$ Hz, 2H), 1.77 – 1.57 (m, 7H), 1.50 – 1.36 (m, 2H), 1.28 (s, 27H), 0.90 (t, $J = 6.8$ Hz, 3H).

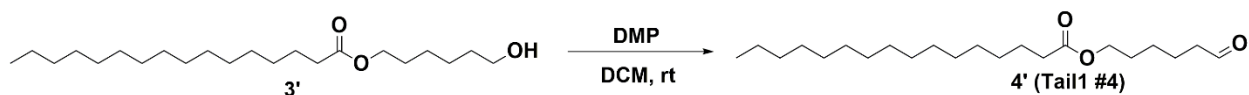
Synthesis of 6-hydroxyhexyl palmitate, **3'**



6-hydroxyhexyl palmitate, **3'** was made using synthesis of 6-hydroxyhexyl oleate, **3'** above.

^1H NMR (500 MHz, CDCl_3) δ 4.09 (t, $J = 6.7$ Hz, 2H), 3.67 (q, $J = 6.2$ Hz, 2H), 2.31 (t, $J = 7.6$ Hz, 2H), 1.64 (tp, $J = 21.2, 6.7$ Hz, 7H), 1.48 – 1.36 (m, $J = 3.6, 3.0$ Hz, 4H), 1.36 – 1.20 (m, 22H), 0.90 (t, $J = 6.9$ Hz, 3H).

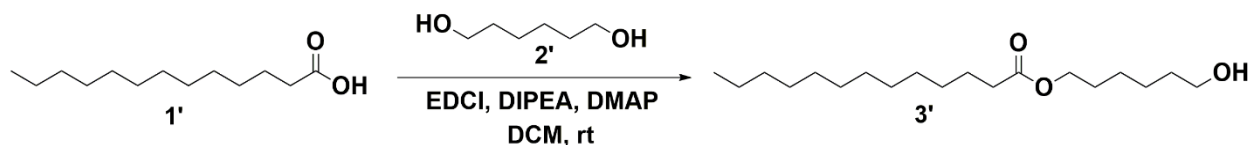
Synthesis of 6-oxohexyl palmitate, 4'



6-oxohexyl palmitate, 4' was made using synthesis of 6-oxohexyl oleate, 4' above.

^1H NMR (500 MHz, CDCl_3) δ 9.79 (s, 1H), 4.09 (t, $J = 6.6$ Hz, 2H), 2.48 (td, $J = 7.3, 1.7$ Hz, 2H), 2.31 (t, $J = 7.5$ Hz, 2H), 1.75 – 1.53 (m, 7H), 1.28 (s, 24H), 0.90 (t, $J = 6.9$ Hz, 3H).

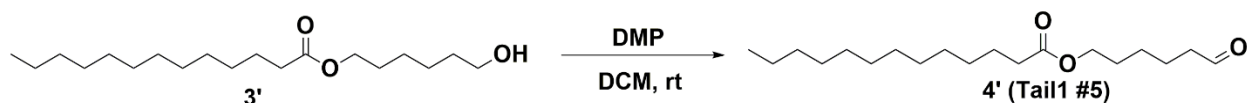
Synthesis of 6-hydroxyhexyl tridecanoate, 3'



6-hydroxyhexyl tridecanoate, 4' was made using synthesis of 6-hydroxyhexyl oleate, 3' above.

^1H NMR (400 MHz, CDCl_3) δ 4.08 (t, $J = 6.7$ Hz, 2H), 3.66 (t, $J = 6.6$ Hz, 2H), 2.30 (t, $J = 7.5$ Hz, 2H), 1.73 – 1.55 (m, 7H), 1.41 (p, $J = 3.6$ Hz, 5H), 1.29 (d, $J = 10.3$ Hz, 18H), 0.90 (t, $J = 6.7$ Hz, 3H).

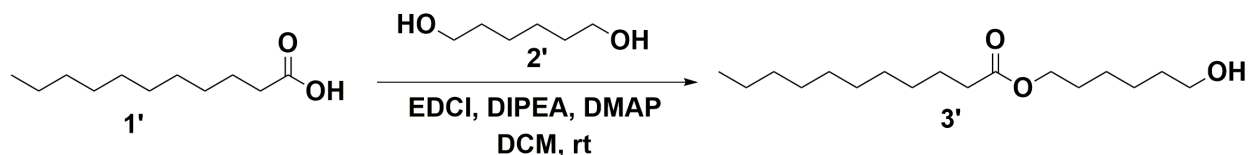
Synthesis of 6-oxohexyl tridecanoate, 4'



6-oxohexyl tridecanoate, 4' (tridecanoic acid) was made using synthesis of 6-oxohexyl oleate, 4' above.

^1H NMR (400 MHz, CDCl_3) δ 9.79 (s, 1H), 4.09 (t, $J = 6.6$ Hz, 2H), 2.47 (td, $J = 7.3, 1.7$ Hz, 2H), 2.31 (t, $J = 7.6$ Hz, 2H), 1.76 – 1.55 (m, 6H), 1.51 – 1.38 (m, 2H), 1.29 (d, $J = 10.1$ Hz, 18H), 0.90 (t, $J = 6.8$ Hz, 3H).

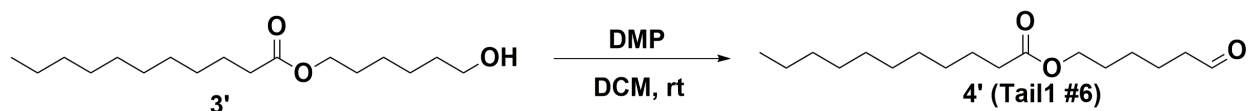
Synthesis of 6-hydroxyhexyl undecanoate, 3'



6-hydroxyhexyl undecanoate, 3' was made using synthesis of 6-hydroxyhexyl oleate, 3' above.

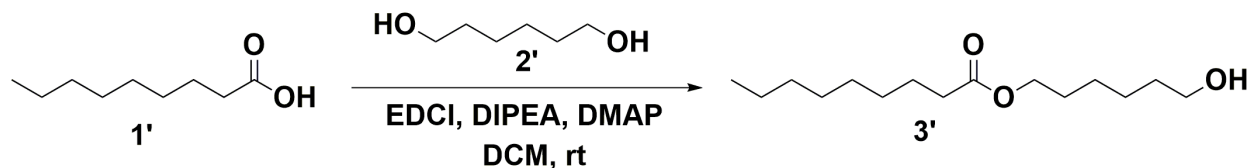
^1H NMR (500 MHz, CDCl_3) δ 4.09 (t, $J = 6.7$ Hz, 2H), 3.67 (t, $J = 6.6$ Hz, 2H), 2.31 (t, $J = 7.5$ Hz, 2H), 1.64 (dp, $J = 22.4, 7.0$ Hz, 6H), 1.42 (dq, $J = 6.7, 2.9$ Hz, 5H), 1.36 – 1.21 (m, 14H), 0.90 (t, $J = 6.9$ Hz, 3H).

Synthesis of 6-oxohexyl undecanoate, 4'



6-oxohexyl undecanoate, 4' was made using synthesis of 6-oxohexyl oleate, 4' above.

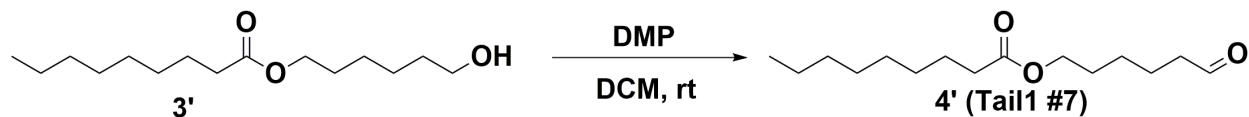
Synthesis of 6-hydroxyhexyl nonanoate, 3'



6-hydroxyhexyl nonanoate, 3' (nonanoic acid) was made using synthesis of 6-hydroxyhexyl oleate, 3' above.

^1H NMR (400 MHz, CDCl_3) δ 4.09 (t, $J = 6.7$ Hz, 2H), 3.67 (t, $J = 6.5$ Hz, 2H), 2.31 (t, $J = 7.6$ Hz, 2H), 1.72 – 1.54 (m, 6H), 1.41 (p, $J = 3.5$ Hz, 5H), 1.37 – 1.20 (m, 10H), 0.90 (t, $J = 6.7$ Hz, 3H).

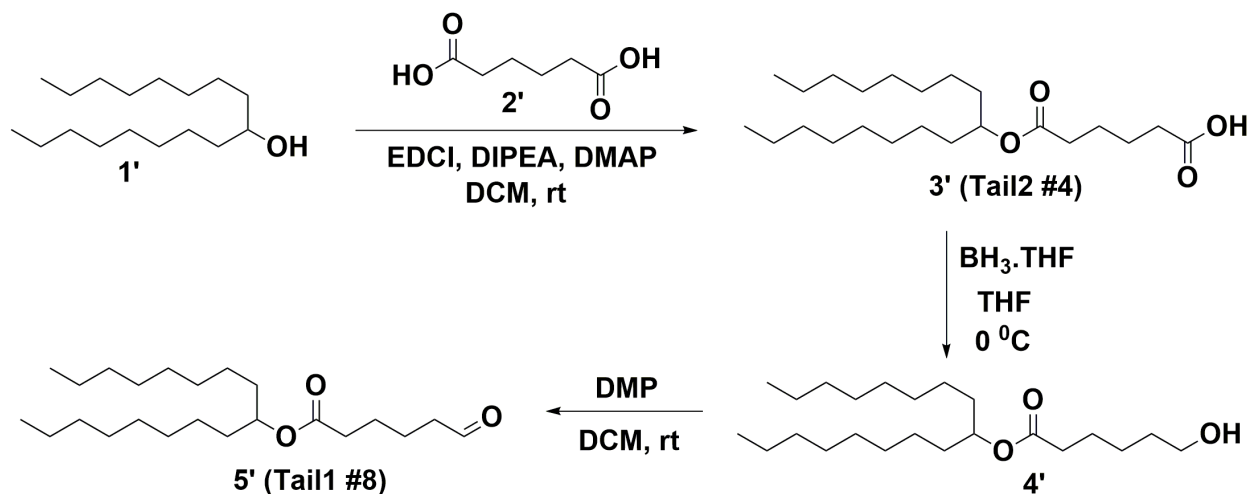
Synthesis of 6-oxohexyl nonanoate, 4'



6-oxohexyl nonanoate, **4'** was made using synthesis of 6-oxohexyl oleate, **4'** above.

^1H NMR (500 MHz, CDCl_3) δ 9.79 (s, 1H), 4.09 (t, $J = 6.6$ Hz, 2H), 2.47 (td, $J = 7.3, 1.7$ Hz, 2H), 2.31 (t, $J = 7.6$ Hz, 2H), 1.75 – 1.57 (m, 7H), 1.46 – 1.37 (m, 2H), 1.34 – 1.23 (m, 9H), 0.90 (t, $J = 6.9$ Hz, 3H).

Synthesis of Tail1 #8



Synthesis of 6-(heptadecan-9-yloxy)-6-oxohexanoic acid, 3' (also Tail2 #4): 9-heptadecanol (3.0 g, 11.7 mmol), *N*-(3-dimethylaminopropyl)-*N'*-ethylcarbodiimide hydrochloride (EDC HCl, 4.5 g, 23.4 mmol), adipic acid (3.4 g, 23.2 mmol), 4-(dimethylamino) pyridine (DMAP, 1.5 g, 11.7 mmol) and *N,N*-diisopropylethylamine (DIPEA, 4.1 ml, 23.4 mmol) were dissolved in dichloromethane (300 ml). The reaction was stirred at room temperature under nitrogen for 18 h. Dichloromethane was evaporated under vacuum, and the residue purified by silica gel chromatography (10% methanol in dichloromethane) to give 6-(heptadecan-9-yloxy)-6-oxohexanoic acid, **3'** (2.3 g, 6.0 mmol, 51 %) as a colorless oil. ^1H NMR (500 MHz, CDCl_3) δ 4.89 (p, $J = 6.3$ Hz, 1H), 2.41 (h, $J = 3.6, 3.2$ Hz, 2H), 2.37 – 2.29 (m, 2H), 1.71 (h, $J = 4.0, 3.5$ Hz, 4H), 1.58 (s, 4H), 1.28 (d, $J = 6.8$ Hz, 24H), 0.90 (t, $J = 6.9$ Hz, 6H).

Synthesis of heptadecan-9-yl 6-hydroxyhexanoate, 4': 6-(heptadecan-9-yloxy)-6-oxohexanoic acid, **3'** (2 g, 5.2 mmol) was dissolved in THF (20 ml) and cooled to 0°C while stirring under nitrogen. BHF.THF (10.5 ml, 10.5mmol) was added dropwise to the reaction, after which the reaction was left to stir at 0°C under nitrogen for 3 hr. The reaction was quenched by adding brine in drops until gas evolution ceased. The mixture was then extracted with ethyl acetate (100 ml). Organic layer was dried over sodium sulfate, filtered, and the organic solvent was evaporated under

vacuum. Crude residue was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give heptadecan-9-yl 6-hydroxyhexanoate **4'** (1.5 g, 4.0 mmol, 77%) as a colorless oil. ¹H NMR (500 MHz, CDCl₃) δ 4.89 (p, *J* = 6.3 Hz, 1H), 3.68 (t, *J* = 6.6 Hz, 2H), 2.33 (t, *J* = 7.5 Hz, 2H), 1.74 – 1.57 (m, 4H), 1.53 (q, *J* = 6.2 Hz, 4H), 1.48 – 1.37 (m, 4H), 1.29 (d, *J* = 8.5 Hz, 23H), 0.90 (t, *J* = 6.8 Hz, 6H).

Synthesis of heptadecan-9-yl 6-oxohexanoate, 5' (Tail #8). Heptadecan-9-yl 6-hydroxyhexanoate, **4'** (1.4 g, 3.8 mmol) was dissolved in dichloromethane 100 ml followed by Dess-Martin Periodinane (2.4 g, 5.7 mmol). The reaction was stirred under nitrogen at room temperature for 2 h. After confirmation of reaction completion by thin layer chromatography, sodium thiosulfate pentahydrate (50% w/v, 200 ml) was added to the reaction and left to stir for additional 15 mins. Organic layer was thereafter separated, washed with brine, dried over Na₂SO₄, filtered, and the filtrate was evaporated under vacuum. The residue was purified by silica gel chromatography (0–50% ethyl acetate in hexanes) to give heptadecan-9-yl 6-oxohexanoate, **5'** (0.8g, 2.2 mmol, 56%) as a light brown oil. ¹H NMR (500 MHz, CDCl₃) δ 9.79 (s, 1H), 4.89 (p, *J* = 6.3 Hz, 1H), 2.49 (tt, *J* = 6.9, 1.7 Hz, 2H), 2.34 (dd, *J* = 10.5, 3.2 Hz, 2H), 1.70 (h, *J* = 4.3, 3.7 Hz, 4H), 1.64 – 1.42 (m, 5H), 1.28 (d, *J* = 6.0 Hz, 23H), 0.90 (t, *J* = 6.8 Hz, 6H).

Synthesis of 7-(((3s,5s,7s)-adamantan-1-yl) amino) – 6 -(N -(3 (dimethylamino) propyl) oleamido)-7-oxoheptyl oleate (119-23). A solution of 6-oxohexyl oleate (0.10g, 0.26 mmol) (Tail 1 #1) and *N*^l, *N*^l-dimethylpropane-1,3-diamine (0.03 g, 0.26 mmol) in dichloromethane (1 ml) in a 4-dram glass vial was stirred at room temperature for 1 hr. Thereafter, oleic acid (0.07g, 0.26 mmol), 1-adamantyl isocyanide (0.04g, 0.26 mmol) and methanol (1 ml) were added. The reaction was left to stir at room temperature overnight. Following confirmation of consumption of starting materials by thin layer chromatography, the reaction mixture was purified by silica gel chromatography (10% methanol in dichloromethane with 0.1% ammonium hydroxide) to give compound **119-23** as a light brown oil (0.12g, 0.13 mmol, 50 %). QTOF MS (ESI): *m/z* calcd for C₅₈H₁₀₆N₃O₄⁺ (M+H), 908.81833; found, 908.8183. ¹H NMR (400 MHz, CDCl₃) δ 6.36 (s, 1H), 5.42 – 5.26 (m, 4H), 4.72 (t, *J* = 7.6 Hz, 1H), 4.05 (q, *J* = 7.0, 6.6 Hz, 2H), 3.31 (ddt, *J* = 28.2, 9.6, 4.8 Hz, 2H), 2.47 – 2.33 (m, 2H), 2.34 – 2.15 (m, 10H), 2.11 – 1.88 (m, 18H), 1.87 – 1.75 (m, 1H), 1.74 – 1.53 (m, 14H), 1.31 (pd, *J* = 11.6, 10.6, 5.9 Hz, 44H), 0.89 (t, *J* = 6.7 Hz, 6H). ¹³C NMR (101 MHz, CDCl₃) δ 174.8, 173.9, 170.5, 130.0, 129.9, 129.8, 129.7, 64.2, 56.9, 51.6, 45.4, 41.5,

36.4, 34.3, 33.5, 31.9, 29.8, 29.7, 29.5, 29.5, 29.4, 29.4, 29.3, 29.2, 29.2, 29.2, 29.1, 28.5, 27.9,
27.9, 27.2, 27.2, 27.2, 26.0, 25.9, 25.7, 25.0, 22.7, 14.1.

QTOF Agilent mass spectra of 119-23

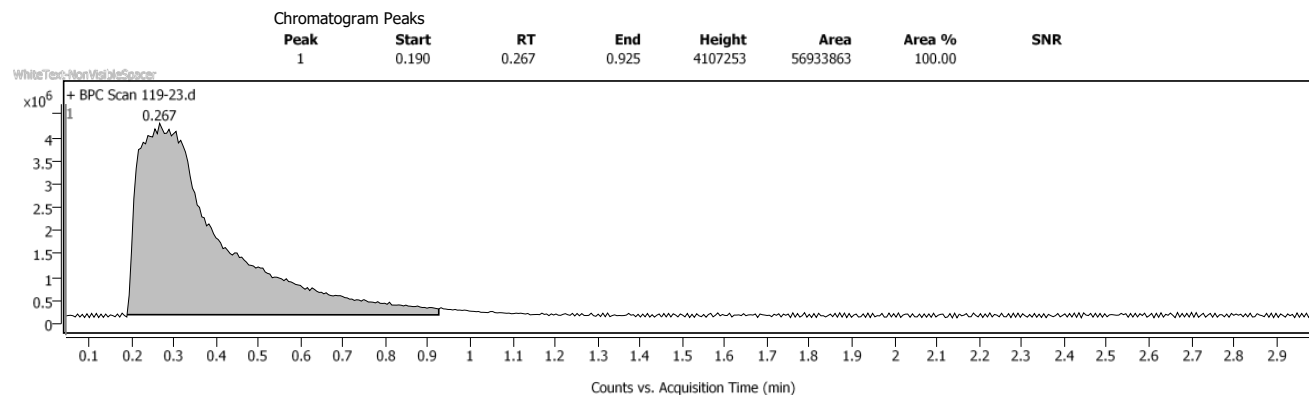
Analysis Report



Sample Information

Name	119-23	Data File Path	D:\MassHunter\Data\Users\Anderson\AEiraj\Repeat117_4CR\119-23.d
Sample ID		Acq. Time (Local)	7/14/2021 6:46:31 PM (UTC-04:00)
Instrument	QTOF	Method Path (Acq)	D:\MassHunter\Methods\Users\Anderson\DI_04182019.m
MS Type	QTOF	Version (Acq SW)	6200 series TOF/6500 series Q-TOF 10.1 (48.0)
Inj. Vol. (ul)	1	IRM Status	Success
Position	P1-A6	Method Path (DA)	D:\MassHunter\Data\Users\Anderson\AEiraj\Repeat117_4CR\119-23.d\AcqData\DI_04182019.m
Plate Pos.		Target Source Path	D:\MassHunter\PCDL\default.csv
Operator		Result Summary	1 qualified (1 targets)

Sample Chromatograms



Sample Spectra

+ Scan (rt: 0.699-0.925 min)

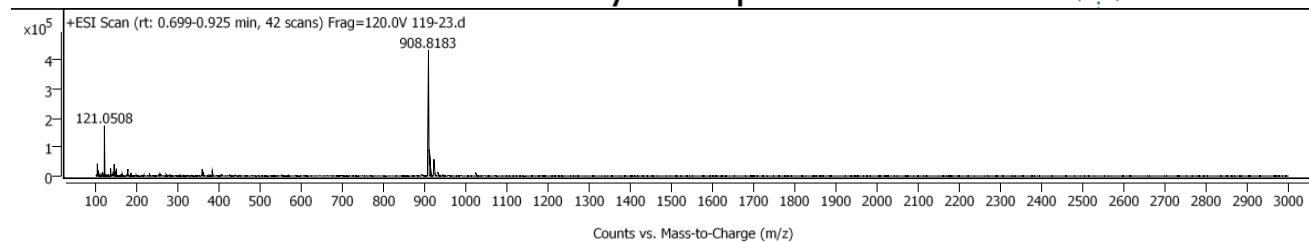
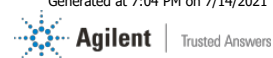
Peak 1 from + BPC Scan

MassHunter Qualitative Analysis

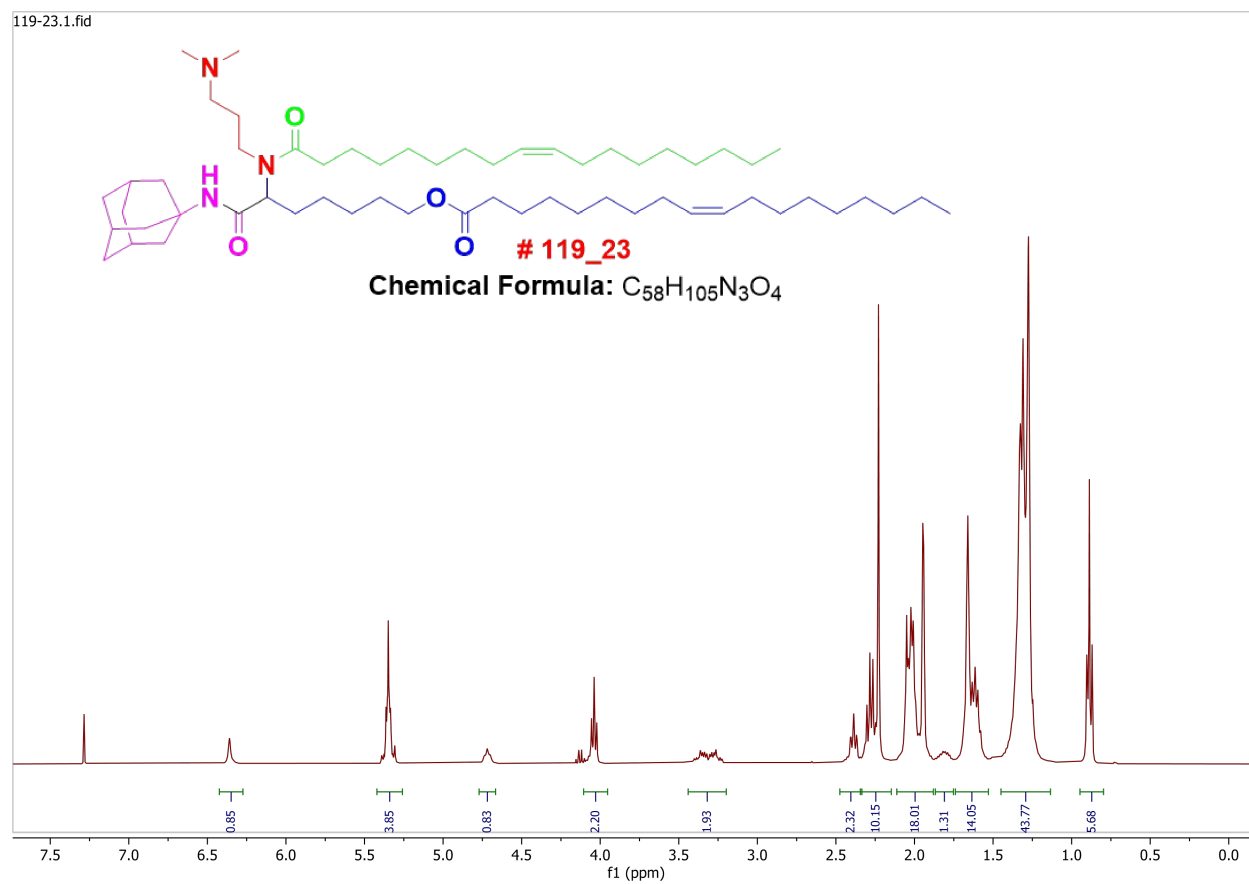
Page 1 of 2

Generated at 7:04 PM on 7/14/2021

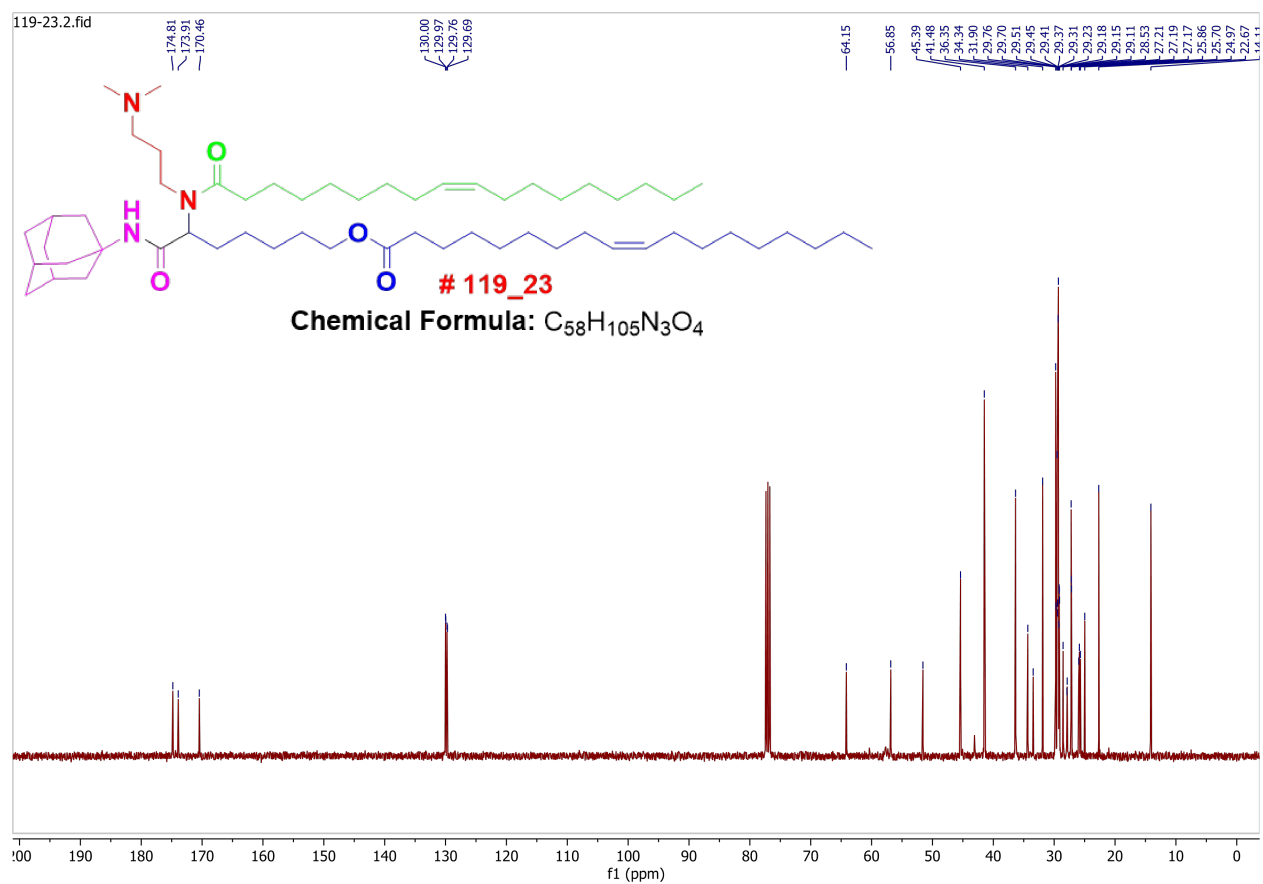
Analysis Report



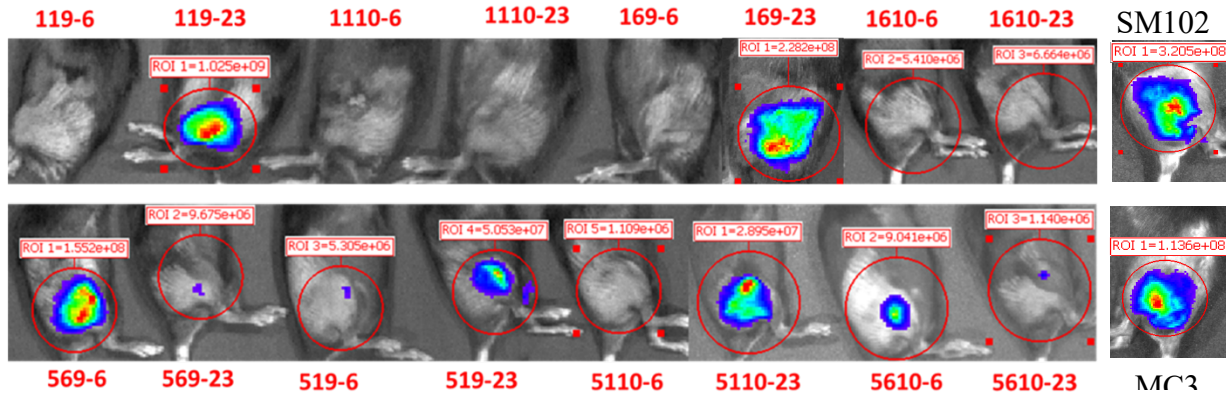
Proton NMR of Lipid 119-23



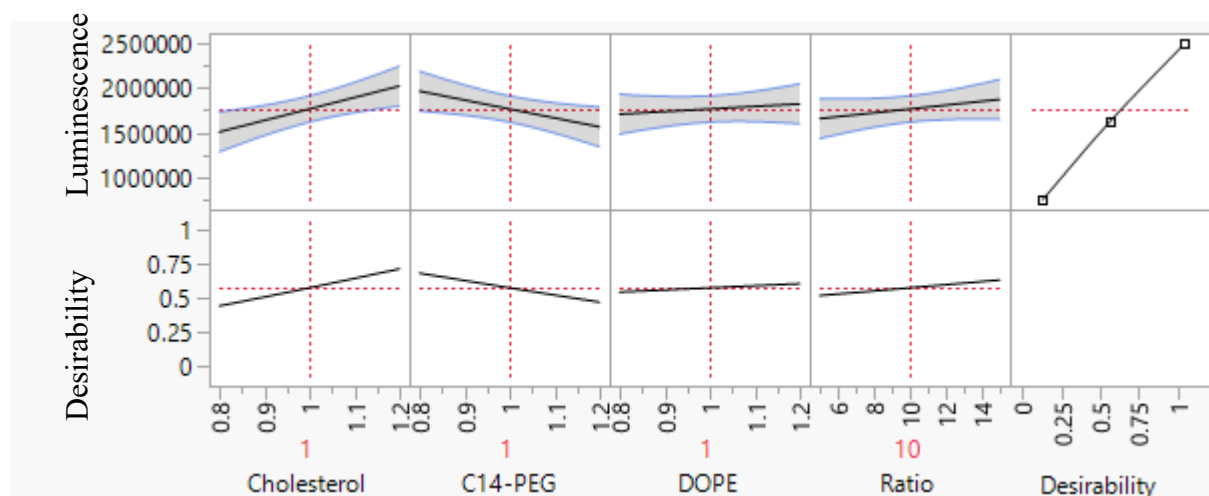
Carbon-13 NMR of Lipid 119-23



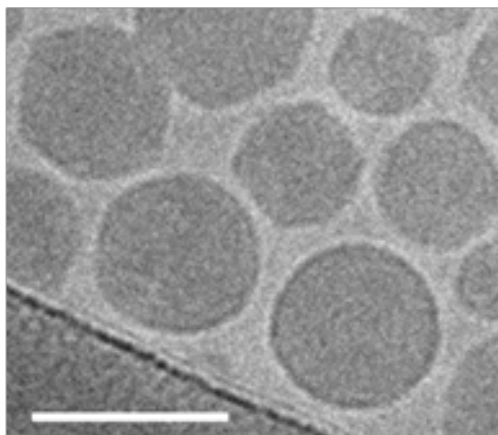
Supplementary Figures



Supplementary Figure 1. Representative IVIS imaging results of luciferase expression 6 h after intramuscular injection of mLuc LNPs made from the 16 ionizable lipids selected by ML (0.1mg/kg mRNA per mouse). mLuc LNPs composed of SM102 and MC3 were tested in parallel as industry-standard benchmarks.



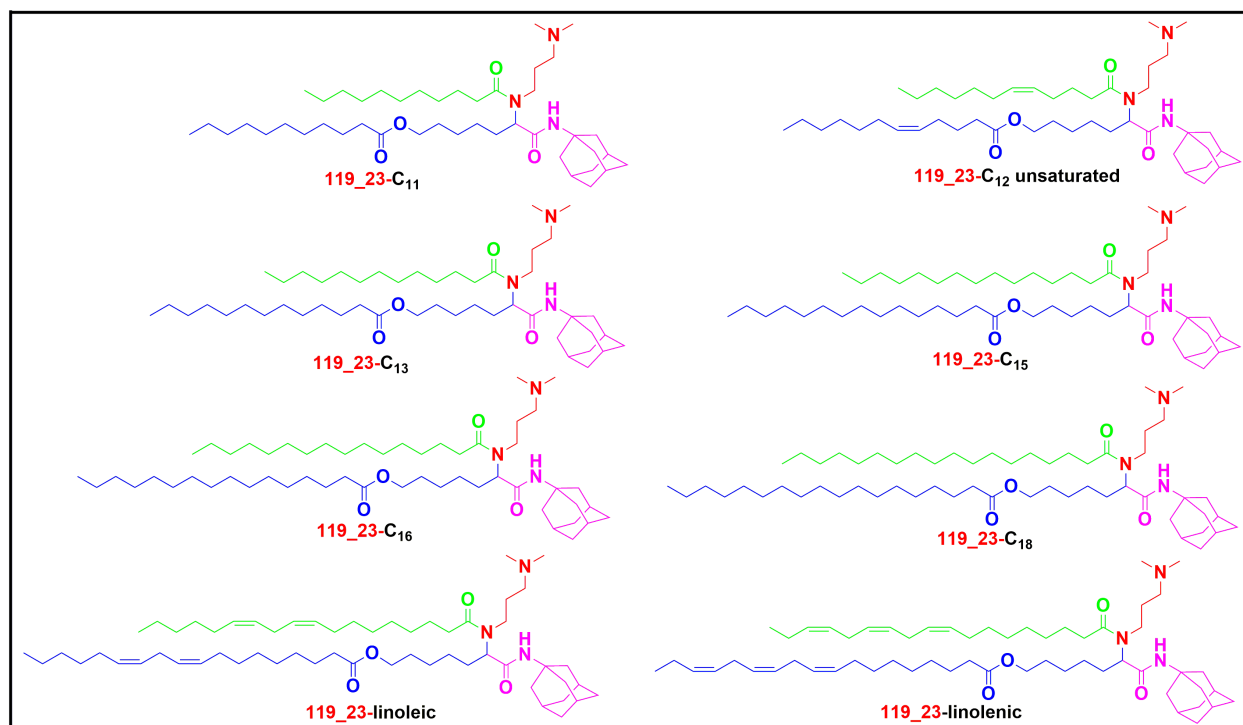
Supplementary Figure 2. Formulation optimization predicted component effects as reported by JMP's Design of Experiment software. Luminescence is as measured at 24 hours following the addition of different LNPs (100ng mLuc per well) to HeLa cells. The predicted effects show that ionizable lipid to mRNA mass ratio ("Ratio") and DOPE content have minimal effects on activity. However, increased cholesterol content and decreased lipid-anchored PEG ("C14-PEG") increase transfection. The shaded regions represent a 95% confidence interval (2 SEM).



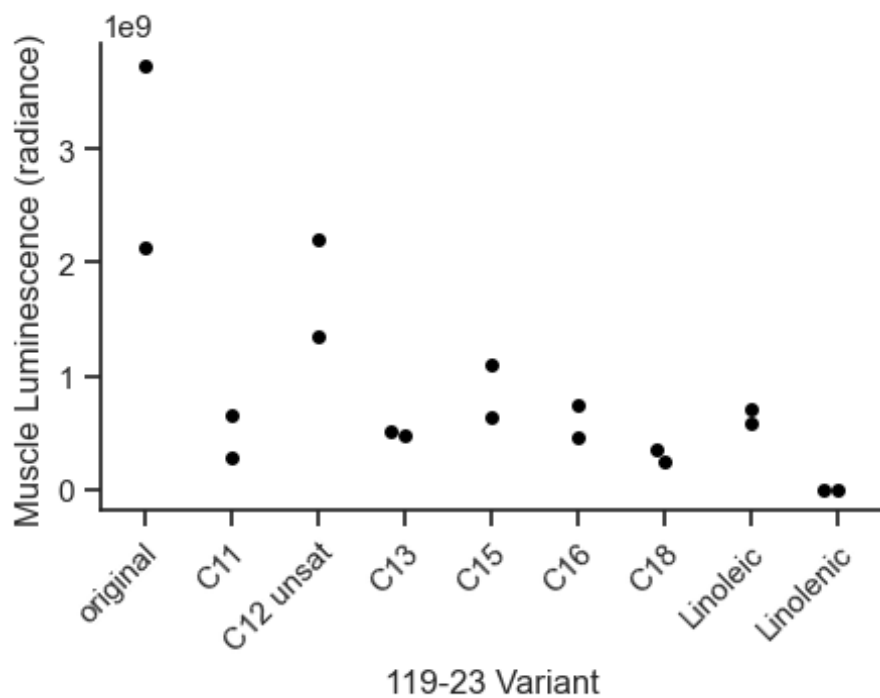
Supplementary Figure 3. TEM image of 119-23 LNP prepared using classic four-component formulations. Scale bar =100nm.



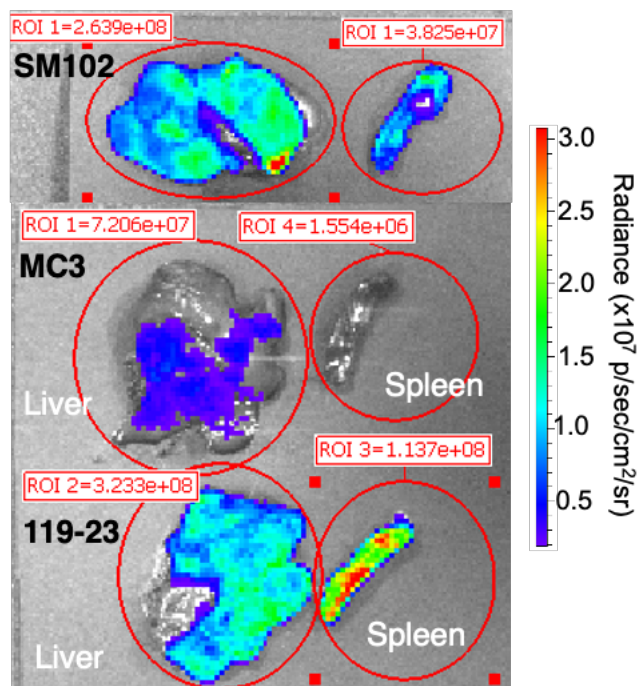
Tail Variants



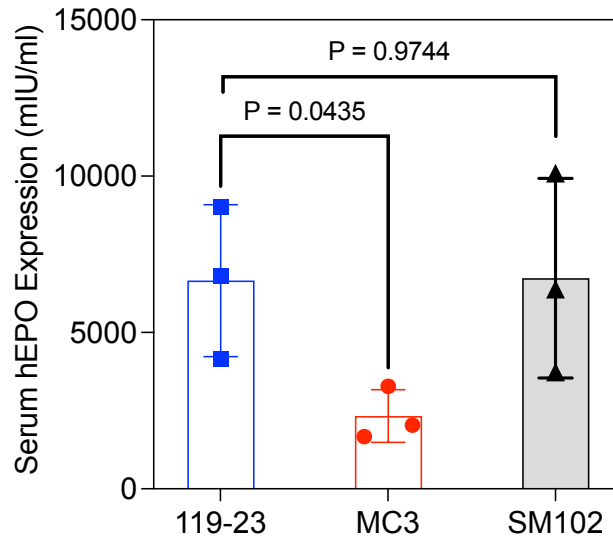
Supplementary Figure 4. Structures of eight 119-23 lipid variants were designed by varying the lipid tail length and the number of unsaturated bonds in lipid tails.



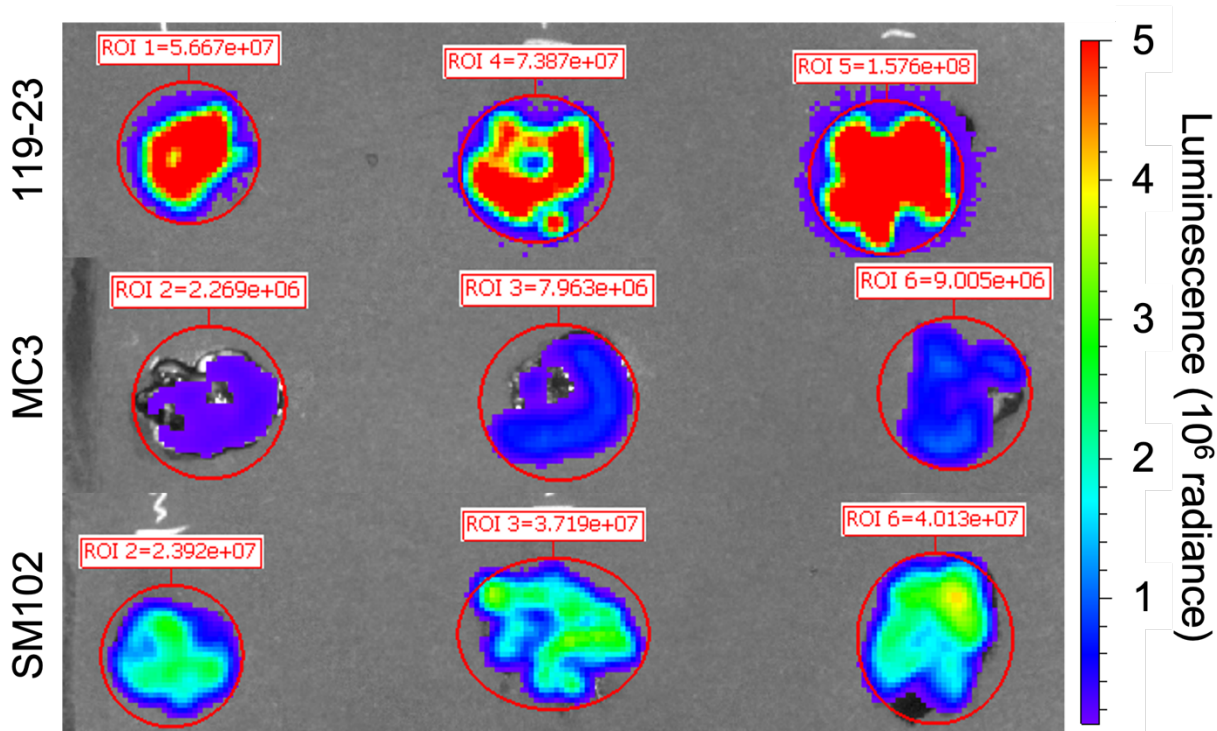
Supplementary Figure 5. The total flux of luciferase signal (ROI) at the injection sites was measured at 6h following intramuscular administration of mLuc LNPs containing 119-23 or its variants after i.m. injection (0.25 mg/kg mRNA per mouse). These LNPs were prepared using classic four-component formulations. Results are presented as the average of n = 2 biologically independent animals.



Supplementary Figure 6. Representative IVIS images of mouse liver and spleen at 6 h following i.v. administration of 119-23, MC3, and SM-102 LNPs at a dosage of 0.25 mg/kg mLuc. These LNPs were prepared using classic four-component formulations.



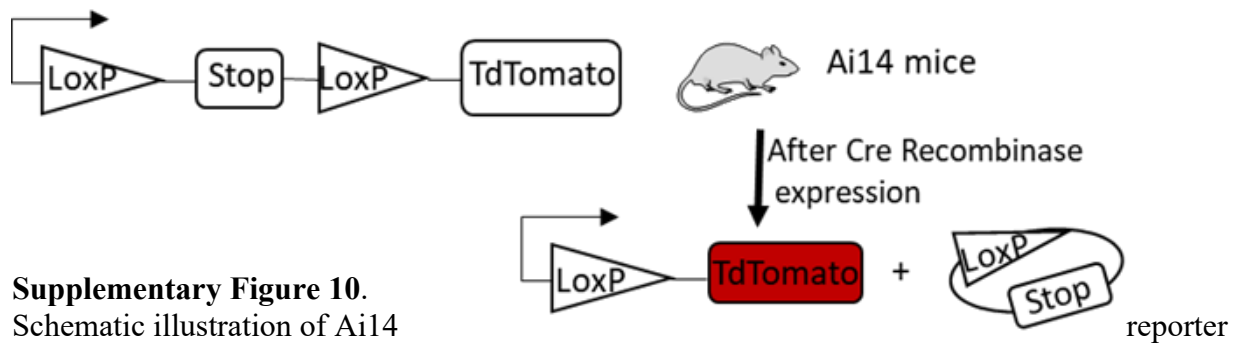
Supplementary Figure 7. hEPO was quantified using an ELISA assay in serum following i.v. administration of 119-23, MC3, and SM-102 LNPs at a dosage of 0.25 mg/kg hEPO mRNA. These LNPs were prepared using classic four-component formulations. Statistical significance was analysed using a one-way analysis of variance (ANOVA) with Dunnett's correction. Results are presented as mean $\pm$ SD with $n = 3$ biologically independent animals.



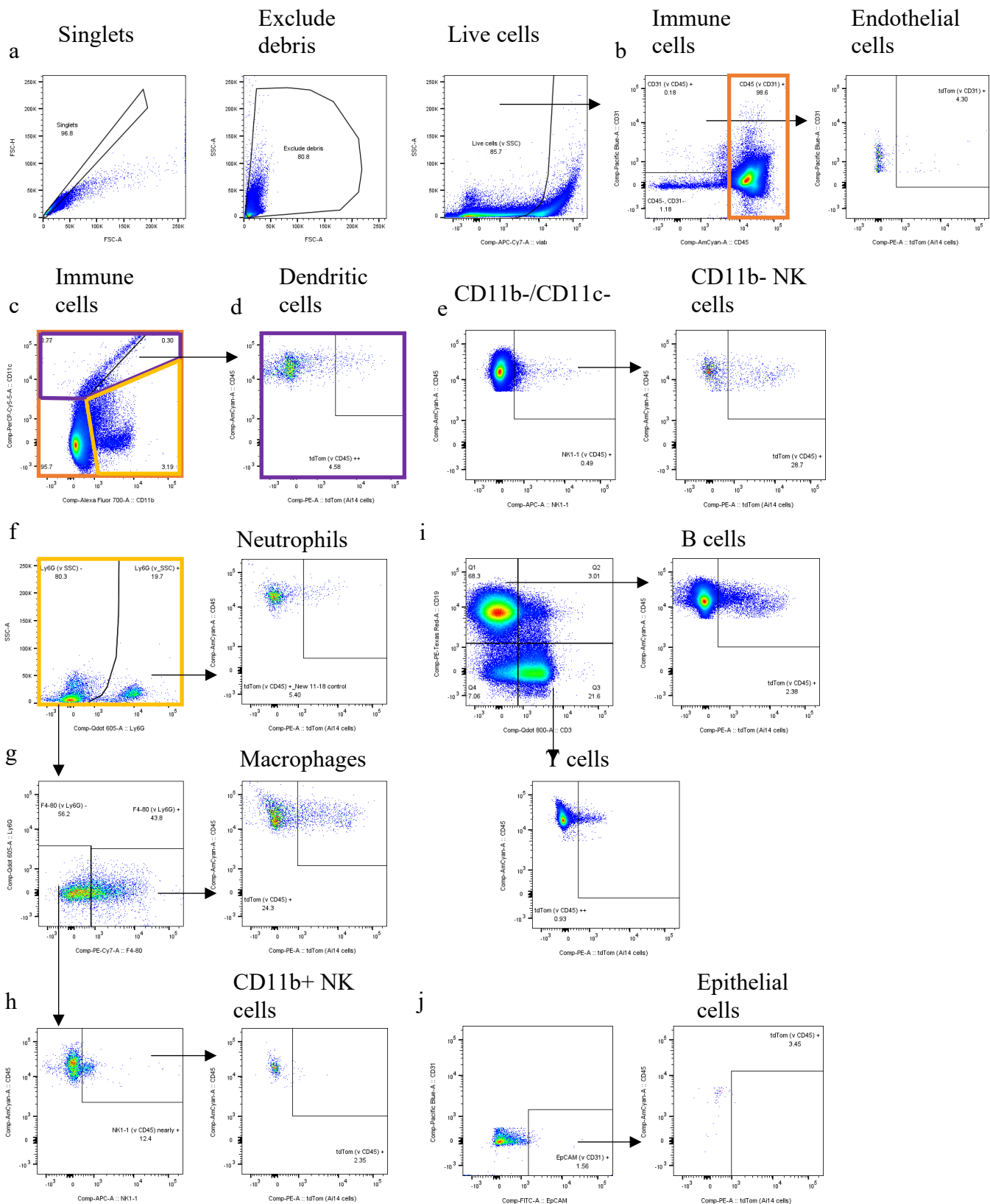
Supplementary Figure 8. Representative IVIS images of mouse lung at 6 h following i.v. administration of 119-23, MC3, and SM-102 LNPs at a dosage of 0.25 mg/kg mLuc. These LNPs were prepared using lung-targeted SORT five-component formulations.

<i>Formulation</i>	Diameter (nm)	PDI	EE%
<i>119-23</i>	95 ± 1.5	0.12 ± 0.02	85.6
<i>MC3</i>	80 ± 1.0	0.10 ± 0.02	95.6
<i>SM102</i>	80 ± 0.50	0.14 ± 0.01	94
<i>119-23 (SORT)</i>	88 ± 0.01	0.22 ± 0.01	86.4
<i>MC3 (SORT)</i>	114 ± 0.60	0.15 ± 0.01	80.2
<i>SM102 (SORT)</i>	95 ± 1.8	0.23 ± 0.01	94.2

Supplementary Figure 9. Characterization of 119-23, MC3, and SM-102 LNPs prepared using classic four-component and lung-targeted SORT five-component formulations (PDI = polydispersity index, EE = encapsulation efficiency).



Supplementary Figure 10.
Schematic illustration of Ai14
mice.



Supplementary Figure 11. Flow cytometry gating strategy. Shown are representative FACS scatter plots, taken from the spleen of a mouse treated with 119-23.

(a) For all treatments and all organs, data were first gated for singlets (FSH-A v FSC-H), then debris (FSC v SSC), and finally on live cells (APC/Cy7 v SSC). **(b)** Live cells (APC/Cy7-) were then gated on CD45 and CD31 expression. Endothelial cells (CD31+, upper left gate) were then gated on PE v CD31 to assess tdTomato expression (in the PE channel).

(c) Immune cells (CD45+, right gate in panel b) were then gated on CD11b and CD11c. **(d)** Dendritic cells (CD45+/ CD11c+, shown in the purple gate) were then gated on PE v CD45 to assess tdTomato expression (in the PE channel). **(e)** The CD11b-/CD11c- population was then gated on NK1.1 v CD45. CD11b- NK cells were given by the NK1.1+ population and then gated on PE v CD45 to assess tdTomato expression (in the PE channel).

(f) Myeloid cells (CD45+/CD11b+/CD11c-, the yellow gate in panels c and e) were then gated on Ly6G v SSC. Neutrophils were given by the Ly6G+ population and then gated on PE v CD45 to assess tdTomato expression (in the PE channel). **(g)** Ly6G- cells were then gated on F4/80 v Ly6G. Macrophages were given by the F4/80+ population and then gated on PE v CD45 to assess tdTomato expression (in the PE channel). **(h)** The F4/80- population was then gated on NK1.1 v CD45. CD11b+ NK cells were given by the NK1.1+ population and then gated on PE v CD45 to assess tdTomato expression (in the PE channel).

(i) The CD11b-/CD11c- population (shown in panel c) was then gated on CD3 v CD19. B cells were given by the CD3-/CD19+ population and then gated on PE v CD45 to assess tdTomato expression (in the PE channel). T cells were given by the CD3+/CD19- population and then gated on PE v CD45 to assess tdTomato expression (in the PE channel).

(j) The CD45-/CD31- population (shown in panel b) was then gated on EpCAM v CD31. Epithelial cells were given by the EpCAM+ population and then gated on PE v CD45 to assess tdTomato expression (in the PE channel).

Source code

```
# %%  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
from statsmodels.tsa.seasonal import seasonal_decompose  
from sklearn.model_selection import train_test_split  
from imblearn.over_sampling import SMOTE  
from xgboost import XGBClassifier  
from sklearn import metrics  
from scipy.stats import chi2_contingency  
  
# %%  
#read data  
data_old= pd.read_excel ('oldLibrary.xlsx')  
data_new= pd.read_excel ('newLibrary.xlsx')  
  
# %%  
#remove col with same values  
data_old=data_old.loc[:, ~data_old.columns.duplicated()]  
data_new=data_new.loc[:, ~data_new.columns.duplicated()]  
#remove cols with all na  
data_old=data_old.dropna(axis=1, how='all')  
data_new=data_new.dropna(axis=1, how='all')  
  
# %%  
#extract features in common  
data_newLibrary=data_new[np.intersect1d(data_new.columns,data_old.columns)]  
data_oldLibrary=data_old[np.intersect1d(data_new.columns,data_old.columns)]  
  
# %%  
#split features by groups  
data_feature_AM_old=data_oldLibrary[data_oldLibrary['Name'].str.contains('amine')]  
data_feature_LK_old=data_oldLibrary[data_oldLibrary['Name'].str.contains('linker')]  
data_feature_TA_old=data_oldLibrary[data_oldLibrary['Name'].str.contains('tailA')]  
data_feature_TB_old=data_oldLibrary[data_oldLibrary['Name'].str.contains('tailB')]  
  
# %%  
#add prefix to column names  
data_feature_AM_old = data_feature_AM_old .add_prefix('AM_')  
data_feature_LK_old = data_feature_LK_old .add_prefix('LK_')  
data_feature_TA_old = data_feature_TA_old .add_prefix('TA_')  
data_feature_TB_old = data_feature_TB_old .add_prefix('TB_')
```

```

# %%
#read experiment results
data_expResult= pd.read_csv ('InVitroResult.csv',index_col=0)

# %%
#merge results with features
data_final= pd.merge(data_expResult, data_feature_AM_old, left_on='amineName',
                     right_on='AM_Name',how='left')
data_final= pd.merge(data_final, data_feature_LK_old, left_on='linkerName',
                     right_on='LK_Name',how='left')
data_final= pd.merge(data_final, data_feature_TA_old, left_on='tailAName',
                     right_on='TA_Name',how='left')
data_final= pd.merge(data_final, data_feature_TB_old, left_on='tailBName',
                     right_on='TB_Name',how='left')

# %%
#binarize results
data_final['binary_result'] = np.where(data_final['result']>=500, 1, 0)
data_final['binary_result']=data_final['binary_result'].astype('int')

# %%
#remove name columns
data_org =
data_final.drop(['amine','tailA','tailB','linker','amineName','tailAName','tailBName','linkerName','
LK_Name','AM_Name','TB_Name','TA_Name'], axis=1)

# %%
#remove columns with all zeros
data_org =data_org .loc[:, (data_org != 0).any(axis=0)]
#remove cols with all NAs
data_org =data_org .dropna(axis=1, how='all')

# %%
#remove cols with all the same values
nunique = data_org .nunique()
cols_to_drop = nunique[nunique == 1].index
data_org =data_org .drop(cols_to_drop, axis=1)

# %%
#section to transform new data into format readable by the model

# %%
#read updated test data
data_new2= pd.read_csv ('newLibrary2.csv')
data_new= pd.read_excel ('newLibrary.xlsx')
data_newLibrary=data_new2

```

```

# %%
#split features by groups
data_feature_AM_new=data_newLibrary[data_newLibrary['Name'].str.contains('amine')]
data_feature_LK_new=data_newLibrary[data_newLibrary['Name'].str.contains('linker')]
data_feature_TA_new=data_newLibrary[data_newLibrary['Name'].str.contains('tailA')]
data_feature_TB_new=data_newLibrary[data_newLibrary['Name'].str.contains('tailB')]

# %%
#add prefix to column names
data_feature_AM_new = data_feature_AM_new.add_prefix('AM_')
data_feature_LK_new = data_feature_LK_new.add_prefix('LK_')
data_feature_TA_new = data_feature_TA_new.add_prefix('TA_')
data_feature_TB_new = data_feature_TB_new.add_prefix('TB_')

# %%
from itertools import product
def expand_grid(dictionary):
    return pd.DataFrame([row for row in product(*dictionary.values())],
                        columns=dictionary.keys())

# %%
#create skeleton df for prediction
dic_candidate={'AM_Name': data_feature_AM_new['AM_Name'],
               'LK_Name': data_feature_LK_new['LK_Name'],
               'TA_Name': data_feature_TA_new['TA_Name'],
               'TB_Name': data_feature_TB_new['TB_Name']}
candidates=expand_grid(dic_candidate)

# %%
#create df for prediction based on new data
data_final_predict=pd.merge(candidates, data_feature_AM_new, on='AM_Name',how='left')
data_final_predict= pd.merge(data_final_predict, data_feature_LK_new,
on='LK_Name',how='left')
data_final_predict= pd.merge(data_final_predict, data_feature_TA_new,
on='TA_Name',how='left')
data_final_predict= pd.merge(data_final_predict, data_feature_TB_new,
on='TB_Name',how='left')

# %%
#remove columns with all zeros
data_final_predict=data_final_predict.loc[:, (data_final_predict != 0).any(axis=0)]
#remove cols with all NAs
data_final_predict=data_final_predict.dropna(axis=1, how='all')

```

```

#extract features in common
data_final_predict=data_final_predict[np.intersect1d(data_final_predict.columns,data_org.columns)]

# %%
#end of section

# %%
#extract features in common
data_final_predict=data_final_predict[np.intersect1d(data_final_predict.columns,data_org.columns)]
data_org=data_org[np.intersect1d(data_final_predict.columns,data_org.columns)]
#assign binary result to data
data_org['binary_result']=data_final['binary_result']

# %%
#assign binary result to training df
data_org['binary_result']=data_final['binary_result']
data_org['binary_result']=data_org['binary_result'].astype(object)

# %%
#check if data is balanced
data_org['binary_result'].mean()

# %%
#split features and target variable
x=data_org[data_org.columns[~data_org.columns.isin(['result','binary_result'])]]
y=data_org['binary_result'].astype('int')

# %%
#rebalance the data by SMOTE, and split training and testing set
oversample = SMOTE()
X, Y= oversample.fit_resample(x, y)
x_train, x_test, y_train, y_test = train_test_split(X, Y, test_size=0.2, random_state=66)

# %%
#training XGBoost model

#clf_xgb = XGBClassifier( booster='gbtree', colsample_bytree=1, gamma=5, max_depth=5,
min_child_weight=3,
#objective='binary:logistic', subsample=1, eta=0.25,use_label_encoder=False)
clf_xgb = XGBClassifier(use_label_encoder=False)
clf_xgb.fit(x_train, y_train)

# %%
#get predicted y for training and testing

```

```

y_pred_test=clf_xgb.predict(x_test)
y_pred_train=clf_xgb.predict(x_train)

# %%
# evaluate model performance -- training set
# accuracy: (tp + tn) / (p + n)
accuracy = metrics.accuracy_score(y_train, y_pred_train)
print('Accuracy: %f % accuracy)
# precision tp / (tp + fp)
precision = metrics.precision_score(y_train, y_pred_train)
print('Precision: %f % precision)
# recall: tp / (tp + fn)
recall = metrics.recall_score(y_train, y_pred_train)
print('Recall: %f % recall)
# f1: 2 tp / (2 tp + fp + fn)
f1 = metrics.f1_score(y_train, y_pred_train)
print('F1 score: %f % f1)
print("AUC:",metrics.roc_auc_score(y_train, y_pred_train))

# %%
# evaluate model performance -- test set
# accuracy: (tp + tn) / (p + n)
accuracy = metrics.accuracy_score(y_test, y_pred_test)
print('Accuracy: %f % accuracy)
# precision tp / (tp + fp)
precision = metrics.precision_score(y_test, y_pred_test)
print('Precision: %f % precision)
# recall: tp / (tp + fn)
recall = metrics.recall_score(y_test, y_pred_test)
print('Recall: %f % recall)
# f1: 2 tp / (2 tp + fp + fn)
f1 = metrics.f1_score(y_test, y_pred_test)
print('F1 score: %f % f1)
print("AUC:",metrics.roc_auc_score(y_test, y_pred_test))

# %%
from random import sample
AM_count=pd.DataFrame()
LK_count=pd.DataFrame()
TA_count=pd.DataFrame()
TB_count=pd.DataFrame()
feature_imp_all=pd.DataFrame()
n_compound=50
n_component=3

for i in range(100):

```

```

randomSeed=sample(range(1000),1)[0]
x_train, x_test, y_train, y_test = train_test_split(X, Y, test_size=0.2,
random_state=randomSeed)
#clf_xgb = XGBClassifier(use_label_encoder=False)
clf_xgb = XGBClassifier( booster='gbtree', colsample_bytree=1, gamma=5, max_depth=5,
min_child_weight=3,
objective='binary:logistic', subsample=1, eta=0.25,use_label_encoder=False)
clf_xgb.fit(x_train, y_train)
feature_imp_temp =
pd.Series(clf_xgb.feature_importances_,index=x_train.columns).sort_values(ascending=False)
feature_imp_all=pd.concat([feature_imp_all,feature_imp_temp])
#predict new data
predicted_result_xgb=candidates
predicted_result_xgb['result']=clf_xgb.predict_proba(data_final_predict)[:,1]
compund_predicted=predicted_result_xgb.sort_values(by='result',ascending=False)

AM=compund_predicted.iloc[0:n_compound,].groupby(['AM_Name']).size().reset_index(name=
'count_AM').sort_values(
by='count_AM',ascending=False).iloc[0:n_component,]
AM_count=pd.concat([AM_count,AM.iloc[:,0]])

LK=compund_predicted.iloc[0:n_compound,].groupby(['LK_Name']).size().reset_index(name=
'count_LK').sort_values(
by='count_LK',ascending=False).iloc[0:n_component,]
LK_count=pd.concat([LK_count,LK.iloc[:,0]])

TA=compund_predicted.iloc[0:n_compound,].groupby(['TA_Name']).size().reset_index(name=
'count_TA').sort_values(
by='count_TA',ascending=False).iloc[0:n_component,]
TA_count=pd.concat([TA_count,TA.iloc[:,0]])

TB=compund_predicted.iloc[0:n_compound,].groupby(['TB_Name']).size().reset_index(name='c
ount_TB').sort_values(
by='count_TB',ascending=False).iloc[0:n_component,]
TB_count=pd.concat([TB_count,TB.iloc[:,0]])
if i%10==0:
    print(i)

# %%
feature_imp_all.reset_index(inplace=True)
feature_imp_all= feature_imp_all.rename(columns = {'index':'Feature',0:'Gain'})

# %%

```

```

importance_sum=feature_imp_all.groupby('Feature')['Gain'].mean().reset_index(name='avg_Gain').sort_values(by='avg_Gain',ascending=False)
importance_sum['group']=importance_sum['Feature'].str[:2]
importance_sum['feature']=importance_sum['Feature'].str[3:]

importance_AM=importance_sum[importance_sum['group']=='AM'].sort_values(by='avg_Gain',ascending=False)
importance_LK=importance_sum[importance_sum['group']=='LK'].sort_values(by='avg_Gain',ascending=False)
importance_TA=importance_sum[importance_sum['group']=='TA'].sort_values(by='avg_Gain',ascending=False)
importance_TB=importance_sum[importance_sum['group']=='TB'].sort_values(by='avg_Gain',ascending=False)

# %%
importance_AM.nlargest(10,['avg_Gain'])

# %%
importance_LK.nlargest(10,['avg_Gain'])

# %%
importance_TA.nlargest(10,['avg_Gain'])

# %%
importance_TB.nlargest(10,['avg_Gain'])

# %%
#get frequency of each component under groups
AM_count=AM_count.rename( columns={0:'AM_Name'} )
LK_count=LK_count.rename( columns={0:'LK_Name'} )
TA_count=TA_count.rename( columns={0:'TA_Name'} )
TB_count=TB_count.rename( columns={0:'TB_Name'} )
AM_Perc=AM_count['AM_Name'].value_counts(normalize=True).reset_index(name='percent')
LK_Perc=LK_count['LK_Name'].value_counts(normalize=True).reset_index(name='percent')
TA_Perc=TA_count['TA_Name'].value_counts(normalize=True).reset_index(name='percent')
TB_Perc=TB_count['TB_Name'].value_counts(normalize=True).reset_index(name='percent')

```